

ACQUIRING DIGITAL DATA;
ETHICS & LAW

THIS LECTURE

Part 1. Web scraping and APIs

- Important concepts and ideas (before actually doing it tomorrow)

Part 2. Law & Ethics of Computational Social Science

- What laws are relevant for CSS research? GDPR
- How do you make sure you're an ethical researcher?
- Practical skills: How to write an IRB request, Data Management Plan, Ethics section?



Part 1: The art of web scraping



WHAT IS WEB SCRAPING?

Using a script to pretend to be a user and automatically collect data from websites.

Transform unstructured web data, typically HTML or JavaScript, into structured data that can be stored and analyzed in a format like CSV, JSON, or a pandas dataframe.

EXAMPLE USES OF WEB SCRAPING

Observational data from websites and platform are among the most important data sources for CSS research:

- Collect social media posts from a platform
- Collecting listings and reviews from Airbnb
- Getting job listings from a job portal
- Getting news paper articles
- Get websites and the links between them
- ... *other examples?*



Search

Advanced search

Less than 5 Proposals X

5 to 10 Proposals X

10 to 15 Proposals X

15 to 20 Proposals X

Clear filters

Category

Select Categories ▾

Experience level

☐ Entry Level (8,708)

☐ Intermediate (80,955)

☐ Expert (30,695)

Job type

☐ Hourly (71,998)

☐ \$ Min /hr \$ Max /hr

☐ Fixed-Price (48,396)

☐ Less than \$100 (22,611)

☐ \$100 to \$500 (17,285)

☐ \$500 - \$1K (4,041)

☐ \$1K - \$5K (3,682)

☐ \$5K+ (776)

Save search

Saved jobs

Sort by: Newest ▾

Posted 59 seconds ago

Looking for Business developer or lead generation expert

Payment unverified ★★★★★ 4.1 \$5 spent PAK

Hourly: \$5.00 - \$6.00 - Expert - Est. time: Less than 1 month, Less than 30 hrs/week

Hello everyone, I am looking for a skilled Lead Generation Expert to identify and deliver high-quality, sales-ready leads. My focus is on clients actively seeking to build custom web solutions, and i need someone who can target decision-makers ready to purchase within this niche. Key...

Social Media Lead Generation Lead Nurturing Lead Generation Content Creation Lead Generation Analysis +6

Posted 1 minute ago

3D Floor Plan Design for Office Space

Payment verified ★★☆☆☆ 1.8 \$6K+ spent Australia

Fixed price - Intermediate - Est. budget: \$200.00

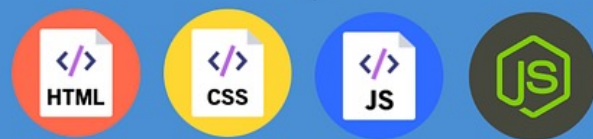
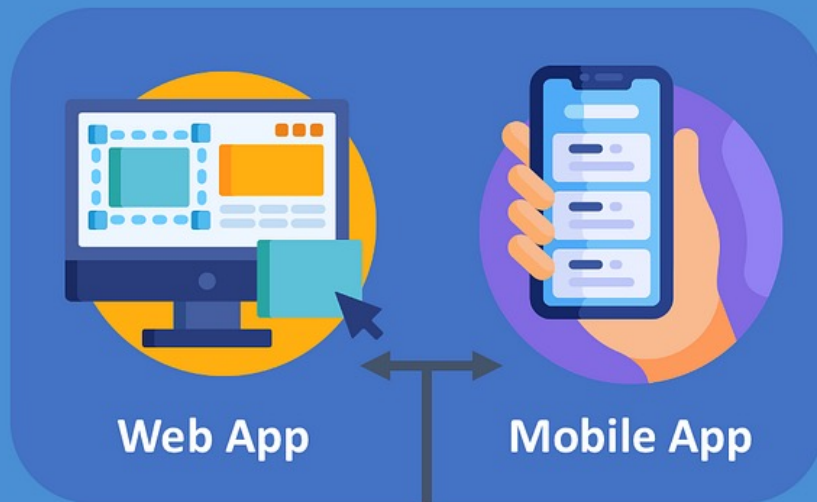
We are looking for a skilled 3D designer to create an accurate floor plan of our office space, including a revised layout featuring furniture placement. The ideal candidate will visualize our requirements and incorporate feedback to produce an appealing and functional design. Experience in...

Interior Design Architectural Design SketchUp Autodesk AutoCAD 3D Rendering

FRONT-END

Anatomy of a website

BACK-END



API



MySQL
PostgreSQL
NoSQL
Mongo



FRONT-END LANGUAGES

This code is being run by your web browser on your computer.

HTML (Hypertext Markup Language): HTML provides the basic structure of the website, using a system of tags and attributes to denote text, links, images, and other content.

CSS (Cascading Style Sheets): CSS is used for styling and visually formatting the HTML elements. It controls layout, colors, fonts, and even some animations.

JavaScript: JavaScript adds interactivity to web pages. It's a scripting language that enables dynamic content, control multimedia, animate images, and much more.



HTML USUALLY HAS OUR CONTENT

HTML is a series of elements or tags that structure content, allowing browsers to display text, links, images, and other resources.

- `<p>This is a paragraph.</p>`
- `<a href="https://www.example.com">This is a link.</a>`
- `<ul><li>This</li><li>is a list</li></ul>`
- `<div class='box'>This is an object of the class 'box'</div>`


```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Simple Website Example</title>
</head>
<body>
<h1>Welcome to Our Simple Website</h1>
<p>This is a demonstration of a simple HTML website designed for parsing practice.</p>
<h2>About Us</h2>
<p>We are a team dedicated to learning web scraping with BeautifulSoup.</p>
<h3>Contact Information</h3>
<p>Email us at: <a href="mailto:info@example.com">info@example.com</a></p>
<h2>Our Products</h2>
<table border="1">
  <tr>
    <th>Product Name</th>
    <th>Description</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Product 1</td>
    <td>An essential item for beginners.</td>
    <td>$19.99</td>
  </tr>
  <tr>
    <td>Product 2</td>
    <td>A must-have for advanced users.</td>
    <td>$29.99</td>
  </tr>
  <tr>
    <td>Product 3</td>
    <td>Now with bacon-flavor!</td>
    <td>$39.99</td>
  </tr>
</table>

</body>
</html>'''
```

Welcome to Our Simple Website

This is a demonstration of a simple HTML website designed for parsing practice.

About Us

We are a team dedicated to learning web scraping with BeautifulSoup.

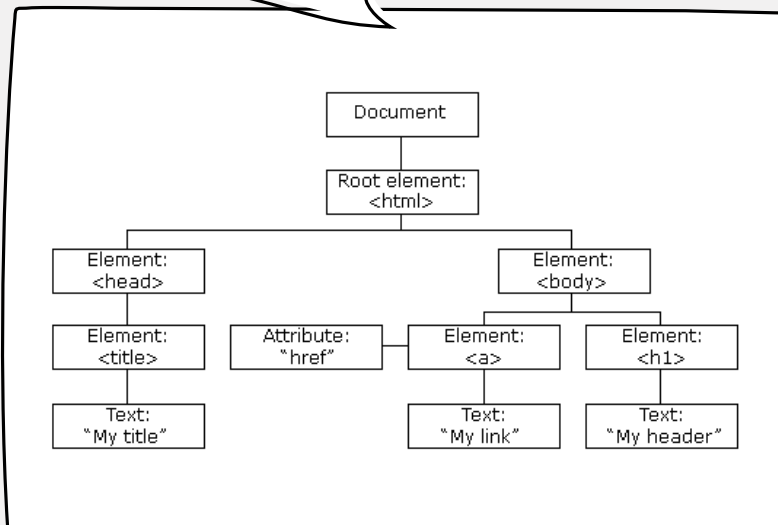
Contact Information

Email us at: info@example.com

Our Products

Product Name	Description	Price
Product 1	An essential item for beginners.	\$19.99
Product 2	A must-have for advanced users.	\$29.99
Product 3	Now with bacon-flavor!	\$39.99

THE DOCUMENT OBJECT MODEL (DOM)



- The DOM is a programming interface for web documents.
- Each element in an HTML document is represented as a node in the DOM tree, making it possible to traverse the hierarchy and access elements programmatically.
- When you use web scraping tools, you're essentially navigating the DOM to access specific parts of the page.



RESEARCH MASTER

Vergelijk

Social Sciences

To solve complex societal issues, we must first investigate them. In this research master's programme, you'll gain diverse perspectives from various social sciences, while developing advanced research skills. This unique combination of interdisciplinary knowledge and research expertise empowers you to explore topics you care deeply about and contribute effectively to innovative solutions.

Language of instruction

English

Mode

Full-time

Location

[Roeterseiland campus](#)

Combination of an interdisciplinary focus and rigorous methodological training

NOV

15

Master's Week: 15-21

Nov

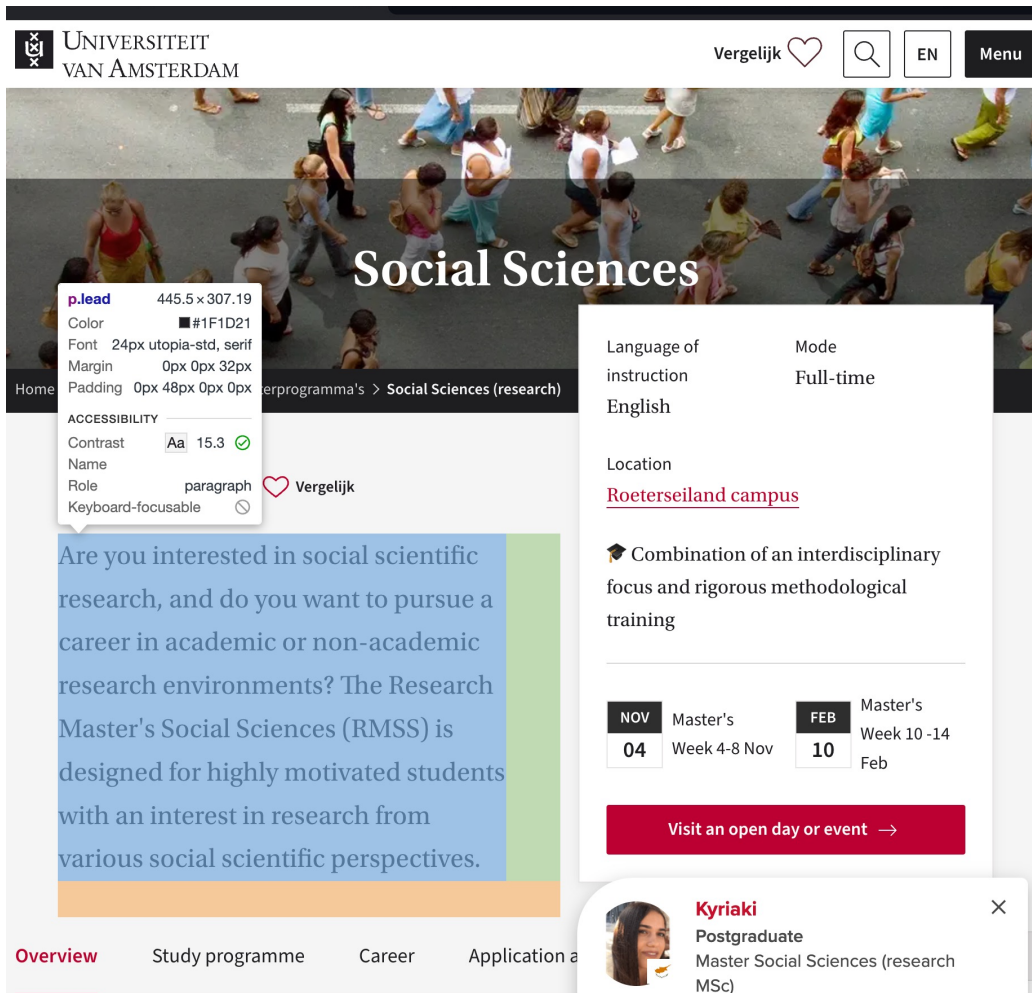
FEB

09

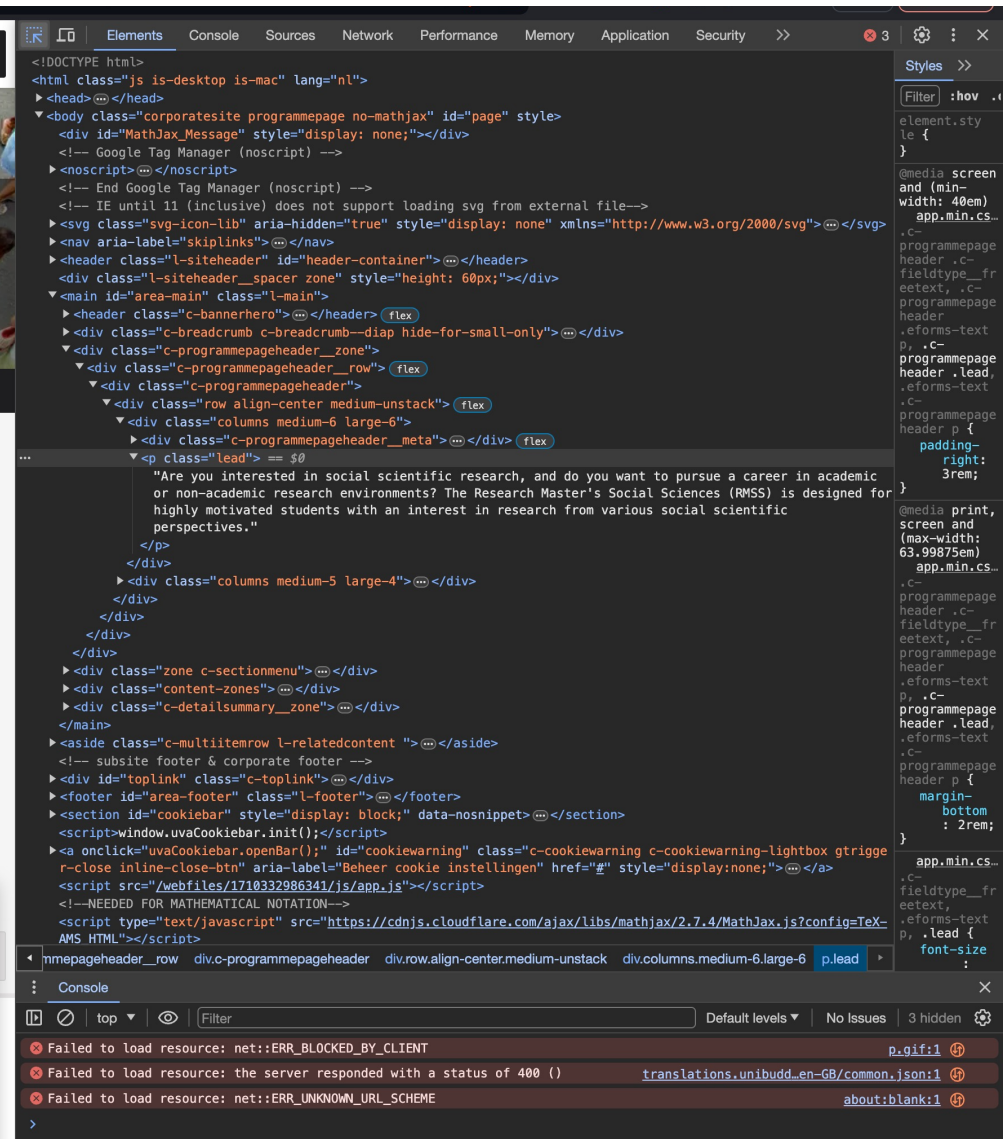
Master's Week 9 -13

Feb

[Visit an open day or event →](#)[Receive the programme summary →](#)[Keep me informed →](#)



Here's why you want to study S
(Research Master's)



PYTHON LIBRARIES FOR SCRAPING

We use programming libraries to scrape:

Requests: simple protocol to communicate with the website and get the HTML

Simple, fast, lightweight and best choice for static pages.

BeautifulSoup: used to easily parse the HTML

Selenium: runs an entire web browser and pretends to be a user clicking around

Slow and heavy, but very capable and works with dynamic websites

EXAMPLE: REQUESTS + BEAUTIFULSOUP

```
import requests

url = "https://example.com"

# Send a GET request to the website
response = requests.get(url)

# Check if the request was successful
if response.status_code == 200:
    html_content = response.text
    print("Page content:")
    print(html_content)
else:
    print(f"Failed to retrieve page. Status code: {response.status_code}")

from bs4 import BeautifulSoup

soup = BeautifulSoup(html_content, 'html.parser')

# Example: Get all links on the page
for link in soup.find_all('a'):
    print(link.get('href'))
```

WEB SCRAPING CHALLENGES

Web scraping can be difficult as websites try to prevent scraping.

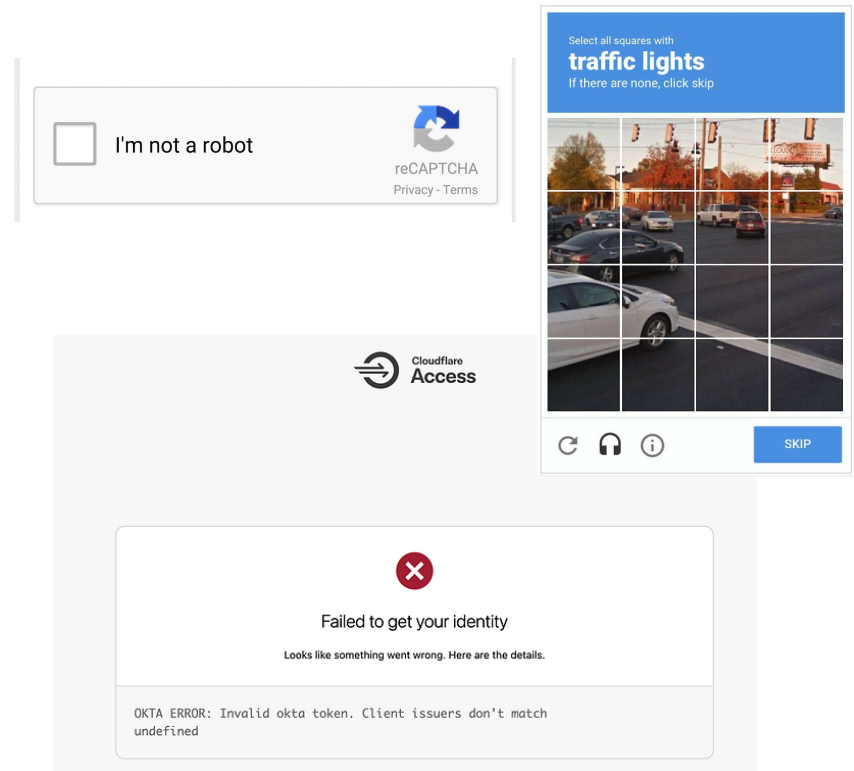
User-agent checks: You have to fake your identity

IP rate limit: Blocks too many calls

Cloudflare: US company. key infrastructure of the internet.
Prevents script access to websites.

Captcha: Designed to prevent scraping.

Handling cookies and sessions: Websites can be complex



ROBOTS.TXT

A document that tells scrapers what the website owner is ok with you scraping or not.

Do you *have* to respect it?

- Legally? No.
- Ethically? No.

But still an important signal!

google.com/robots.txt:

User-agent: *

Disallow: /search

Allow: /search/about

Allow: /search/static

Allow: /search/howsearchworks

Disallow: /sdch

Disallow: /groups

Disallow: /index.html?

Disallow: /?

Allow: /?hl=

Disallow: /?hl=*&

Allow: /?hl=*&gws_rd=ssl\$

Disallow: /?hl=*&*&gws_rd=ssl

Allow: /?gws_rd=ssl\$

Allow: /?pt1=true\$

Stormfront > General > History & Revisionism

The Hitler We Loved And Why

Donate

Register

Blogs

History & Revisionism Reexamining history, particularly the

Post Reply

10-16-2023, 11:07 PM

Gott Mit Uns

Alter Kämpfer
["Friend of Stormfront"](#)
[Sustaining Member](#)

Join Date: Jan 2023
 Posts: 892
 ■ ■ ■

The Hitler We Loved And Why

The Hitler We Loved and Why by Eric Thomson and Christof
[The Hitler We Loved and Why](#)

PDF:
<https://der-fuehrer.org/bucher/engli...0and%20Why.pdf>

Video:
[The Hitler We Loved and Why by Eric Thompson](#)

EXAMPLE: STORMFRONT.ORG

- Long-standing large Nazi community
- How do we understand 'echo chambers'? What actually takes place within them?
- Scraped all comments and discussions

Every month is
 White history month

User Name ☐ Remember Me
 Password

Day's Posts

Search



PETTERTORNBORG · UPDATED 2 YEARS AGO



1

New Notebook

Download (2 GB)



Stormfront.org - White Power forum posts 2001-2020

Hate speech from White Supremacists: 10M posts over nearly 20 years

Data Card

Code (0)

Discussion (0)

Suggestions (0)



About Dataset

Stormfront.org is a white supremacist forum that has been active for 20 years, making it one of the longest-running forums on the Internet. This dataset contains the time-stamped text from all 10M posts made on Stormfront in the 2001-2020 period. The data can for instance be used to track the evolution of language as users engage with a far-right community. Each post is tagged by language (automatically identified). Usernames are anonymized, but a unique id is provided.

The file can be read from Python by:

```
df = pd.read_csv('stormfrontposts.csv.gz', compression='gzip')
```

More information about the dataset can be found in the following article:

Törnberg, P. Törnberg, A. 2022. Inside a White Power Echo Chamber: Why Fringe Digital Spaces are Polarizing Politics. *New Media & Society*.

If you use any of the provided material in your work, please cite us as follows:

Usability ⓘ

7.06

License

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Expected update frequency

Never

Tags

Online Communities

Text

People and Society

API (APPLICATION PROGRAMMING INTERFACE) DATA

- APIs are made to facilitate data exchange between applications.
- APIs send structured data, such as JSON, rather than HTML.
- Many platforms provide public API access, which allows you to get data.
- This data is very easy to parse programmatically:
`parsed_json = json.loads(json_string)`
- APIs come with manuals, and often have libraries for making it even easier! E.g. Tweepy for Twitter.
- Research data from Twitter, Reddit, Facebook, etc., generally came from their APIs

```
{
  "longitude": 47.60,
  "latitude": 122.33,
  "forecasts": [
    {
      "date": "2015-09-01",
      "description": "sunny",
      "maxTemp": 22,
      "minTemp": 20,
      "windSpeed": 12,
      "danger": false
    },
    {
      "date": "2015-09-02",
      "description": "overcast",
      "maxTemp": 21,
      "minTemp": 17,
      "windSpeed": 15,
      "danger": false
    },
    {
      "date": "2015-09-03",
      "description": "raining",
      "maxTemp": 20,
      "minTemp": 18,
      "windSpeed": 13,
      "danger": false
    }
  ]
}
```

SOCIAL SCIENCE ANALYSIS AS A SERVICES

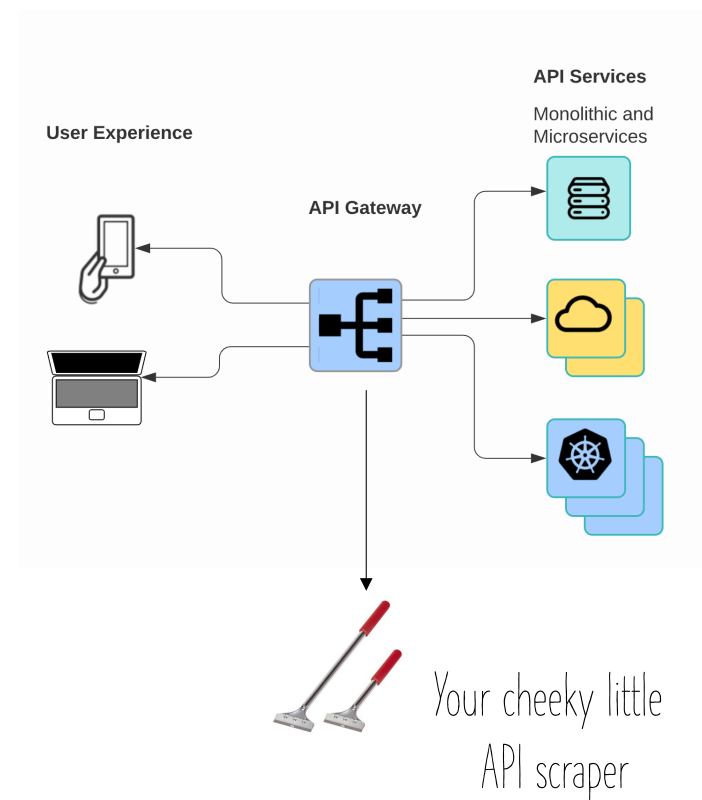
APIs also provide access to services: they allow code to interact with an online service.

- MTurk is based on an API: humans-as-a-service
- Platform LLMs (Claude, ChatGPT, etc) are available through API. So are almost all LLMs.
- Many AI, NLP and image analysis tools are available through APIs
- We will use Perspective Toxicity API to analyze how toxic a comment is

GETTING DATA FROM INTERNAL APIS

- Many dynamic websites and apps run on internal APIs
- Can also be a strategy for scraping: direct access to internal APIs by reverse engineering

For instance, Instagram!




```
import requests

cookies = {
    'datr': 'quu8Z4BY8Xtm8hnGltQ2EwJY',
    'csrftoken': 'PgJBHugGIGK8zth9uHtmGz650zDoLov',
    'ig_did': '90763CE2-2BDC-4B44-A142-217346EE0F52',
    'mid': 'Z_0DdgAAAE12PE6pa85x2l9vwUi',
    'sessionId': '192593382%3ArqZ5oQ7YmfEd5P%3A6%3AAyEYDI3ZpQd0fEUwLmOLZtZECEx8G6hFeHphybzH0w',
    'ds_user_id': '192593382',
    'rur': 'FRC\\054192593382\\0541775548178:01f7ab0ca5ee20cfb838469cd56a7c796c342baab21299379f4a05de64bb0f6733ed3c5d',
    'wd': '857x1045',
}

headers = {
    'accept': '*/',
    'accept-language': 'en-US,en;q=0.8',
    'content-type': 'application/x-www-form-urlencoded',
    'origin': 'https://www.instagram.com',
    'priority': 'u=1, i',
    'referer': 'https://www.instagram.com/',
    'sec-ch-ua': '"Chromium";v="134"',
    'sec-ch-ua-full-version-list': '"Chromium";v="134.0.0.0", "Not:A-Brand";v="24.0.0.0", "Brave";v="134.0.0.0"',
    'sec-ch-ua-mobile': '?0',
    'sec-ch-ua-model': '""',
    'sec-ch-ua-platform': 'macOS',
    'sec-ch-ua-platform-version': '14.6.1',
    'sec-fetch-dest': 'empty',
    'sec-fetch-mode': 'cors',
    'sec-fetch-site': 'same-origin',
    'sec-gpc': '1',
    'user-agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36',
    'x-asbd-id': '359341',
    'x-bloks-version-id': '0d99de0d13662a50e0958bcb112dd651f70dea02e1859073ab25f8f2af477de96',
    'x-csrftoken': 'PgJBHugGIGK8zth9uHtmGz650zDoLov',
    'x-fb-friendly-name': 'PolarisProfilePostsQuery',
    'x-fb-ld': 'BY3c_od_cnaQhTpn06wdYr',
    'x-ig-app-id': '936619743392459',
    # 'cookie': 'datr=quu8Z4BY8Xtm8hnGltQ2EwJY; csrftoken=PgJBHugGIGK8zth9uHtmGz650zDoLov; ig_did=90763CE2-2BDC-4B44-A142-217346EE0F52; mid=Z_0DdgAAAE12PE6pa85x2l9vwUi; sessionId=192593382%3ArqZ5oQ7YmfEd5P%3A6%3AAyEYDI3ZpQd0fEUwLmOLZtZECEx8G6hFeHphybzH0w; ds_user_id=192593382; rur=FRC\\054192593382\\0541775548178:01f7ab0ca5ee20cfb838469cd56a7c796c342baab21299379f4a05de64bb0f6733ed3c5d; wd=857x1045',
}

data = {
    'av': '17841400521463009',
    'd': 'www',
    'user': '0',
    'a': '1',
    'req': '1o',
    'hs': '20185.HYP:instagram_web_pkg.2.1...1',
    'dpr': '2',
    'ccg': 'GOOD',
    'rev': '1021627382',
    's': 'io8b45vxe68p:kwfz00',
    'hsi': '7490475217097250447',
    'dyn': '7xelUjG1mxuIsyaxG4Vp41twpUnwgU7SbzEdF8aUco2qwJyEiw9-1DwUx60p-0LVE4W0qa321Rw8G11wbz81s8hwGxa3a1YwBgao6C0M02iyo7u31fK0zEkxe2GewGw9a3614xm0zK5o4q3y261kx-0ma2-azqwt8d-2u2J080321LwTwKG1pg2fwxyo601FwIA3a3zhA6bwIDyXxuiZqiUqwm8jxK2K0P8K9x6',
    'csr': 'ggMlqgs0a0HAJhi8yrsticz0BSPuYZlt4W0Gj0bCxXnrASApvF7HiGiJ69heT8nAOIF6t2av96q8AJY8G6VqkEiihXnyXBDIGkxKaIBrFfAAAZmpUgpykiQIQF_BADyFfzKWBWBtpdp9pu8TUKV-LUK5C4d5GbyoF2F4p95ByV8DAxyqimUtiK4EB2qCxyXGczqCw04WHa0vKq7FE10pEhYE6W1L8eWAgCdGv7PCwoEronyXx21e8EdEy2a2lq8m6oze0mmzjG2q2ubJ05Tw30oS1_UKfU5q0gy09140Mw6Q4Ja37xKaEE6B1Sfg2ACoAj81kyV83Hw6ww4fw4xy9no4d122m7y3410wnCq0k04mw5HvzFx4g1AwP49wHg2axe6Uo8q1uyk1LixC0TUdoaK5m5Q5UdEy01pDtAghxu3y04BU-4omw0x22m32lo0b9m084w',
    'comet_req': '7',
    'fb_dtsg': 'NACnt_Yo3qjwkvXe0oBbysHwJxpv2-03M7F4cb7sUCahL9VbvnUHya:17865379441060568:1744012156',
    'jazoest': '26528',
    'lsd': 'BY3c_od_cnaQhTpn06wdYr',
    'spin_r': '1021627382',
    'spin_b': 'trunk',
    'spin_t': '1744012166',
    'crn': 'comet.igweb.PolarisFeedRoute',
    'fb_api_caller_class': 'RelayModern',
    'fb_api_req_friendly_name': 'PolarisProfilePostsQuery',
    'variables': '{"data":{"count":12,"include_reel_media_seen_timestamp":true,"include_relationship_info":true,"latest_besties_reel_media":true,"latest_reel_media":true},"username":"afd.bund","__relay_internal__pv__PolarisIsLoggedInRelayprovider":true,"__relay_internal__pv__PolarisShareSheetV3Relayprovider":true}',
    'server_timestamps': 'true',
    'doc_id': '9750506811647048',
}

response = requests.post('https://www.instagram.com/graphql/query', cookies=cookies, headers=headers, data=data)
```

AI AUDITING: LLM OUTPUT AS DATA

We will use LLMs as a tool through API (next week!)

But APIs can also allow us to study LLM behavior

We run 1000s of calls to the LLM and see how it responds.

Political attitudes and framing → How do models describe political actors, policies, or conflicts?

Cross-linguistic variation → Are outputs consistent across languages and cultural contexts?

Reliability and factuality → When do models hallucinate or produce misleading claims?

Normative alignment → How do models encode values (e.g. fairness, harm, safety)?

EXAMPLE STUDY

Provide description of a person and ask LLM to extract information.

Vary only name and accent.

Result: Black, Hispanic, Asian sounding names are consistently seen as less educated, professional etc.

Large Language Models Reproduce Racial Stereotypes When Used for Text Annotation

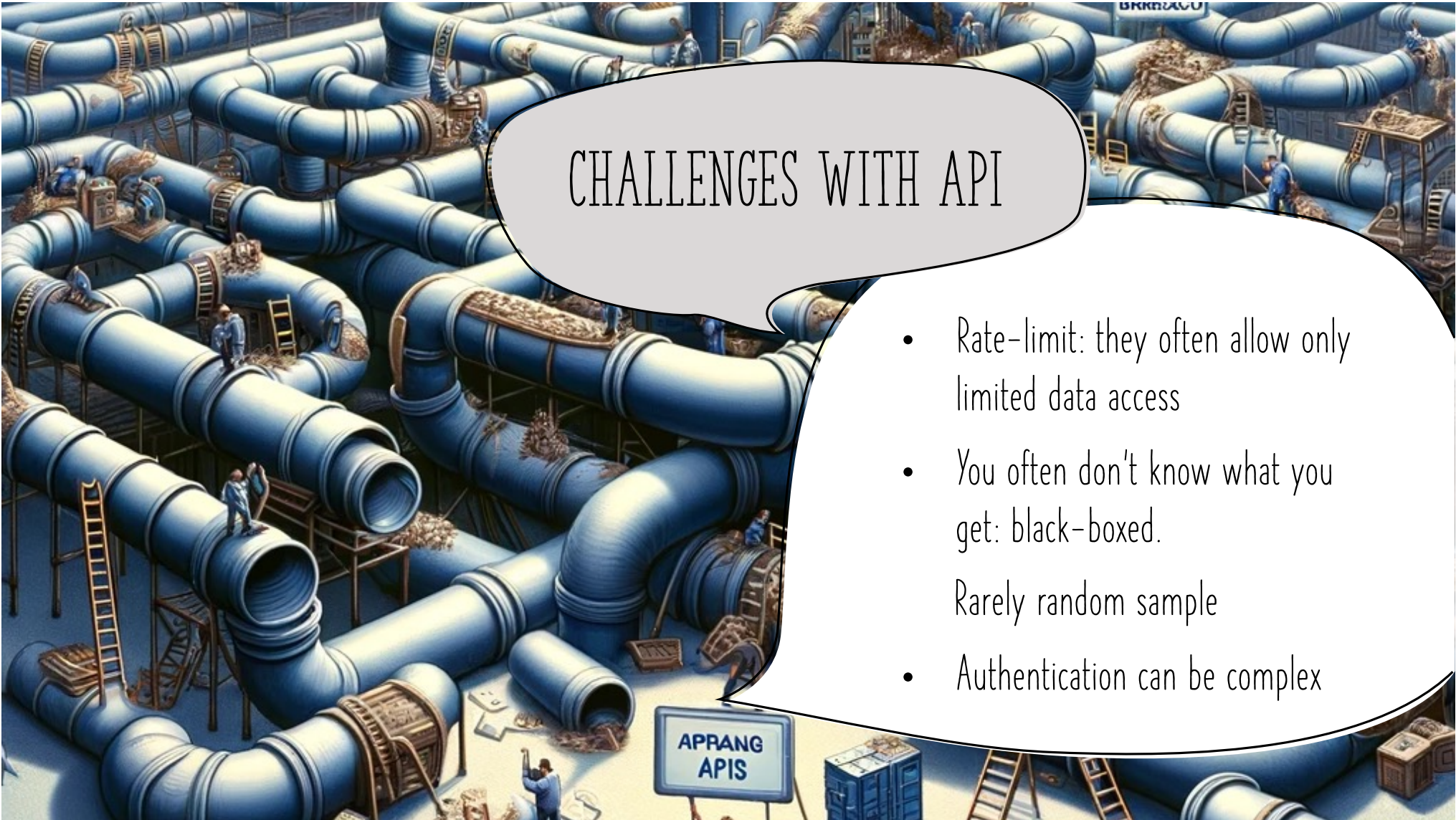
Petter Törnberg¹

¹Institute for Logic, Language and Computation (ILLC), University of Amsterdam, Science Park 107, 1098 XG Amsterdam, Netherlands

Abstract

Large language models (LLMs) are increasingly used for automated text annotation in tasks ranging from academic research to content moderation and hiring. Across 19 LLMs and two experiments totaling more than 4 million annotation judgments, we show that subtle identity cues embedded in text systematically bias annotation outcomes in ways that mirror racial stereotypes. In a names-based experiment spanning 39 annotation tasks, texts containing names associated with Black individuals are rated as more aggressive by 18 of 19 models and more gossipy by 18 of 19. Asian names produce a bamboo-ceiling profile: 17 of 19 models rate individuals as more intelligent, while 18 of 19 rate them as less confident and less sociable. Arab names elicit cognitive elevation alongside interpersonal devaluation, and all four minority groups are consistently rated as less self-disciplined. In a matched dialect experiment, the same sentence is judged significantly less professional (all 19 models, mean gap -0.774), less indicative of an educated speaker (-0.688), more toxic (18/19), and more angry (19/19) when written in African American Vernacular English rather than Standard American English. A notable exception occurs for name-based hireability, where fine-tuning appears to overcorrect, systematically favoring minority-named applicants. These findings suggest that using LLMs as automated annotators can embed socially patterned biases directly into the datasets and measurements that increasingly underpin research, governance, and decision-making.

Keywords: large language models; text annotation; racial bias; stereotype; African American Vernacular English; algorithmic discrimination



CHALLENGES WITH API

- Rate-limit: they often allow only limited data access
- You often don't know what you get: black-boxed.
Rarely random sample
- Authentication can be complex



THE APICALYPSE

- Many APIs have been shutting down
- Instagram, Reddit, Twitter, Facebook...
- In part due to costs, privacy concerns, and in part to LLMs
- But DSA might help: an important new EU route for qualified/vetted researchers to access data from very large platforms, especially for systemic-risk research, but it is not a general replacement for open APIs.

RESPONSIBLE DATA COLLECTION PRACTICES

- When scraping or collecting data from an APIs, don't collect too quickly since it can overburden the server. Pause between requests e.g., *time.sleep(0.5)*
- Don't collect data on people that you don't need (data minimization)
- Be mindful of privacy and ethical implications
- Consider anonymizing the data before storing (anonymization)

SHOULD YOU SCRAPE AS PART OF YOUR PROJECT?

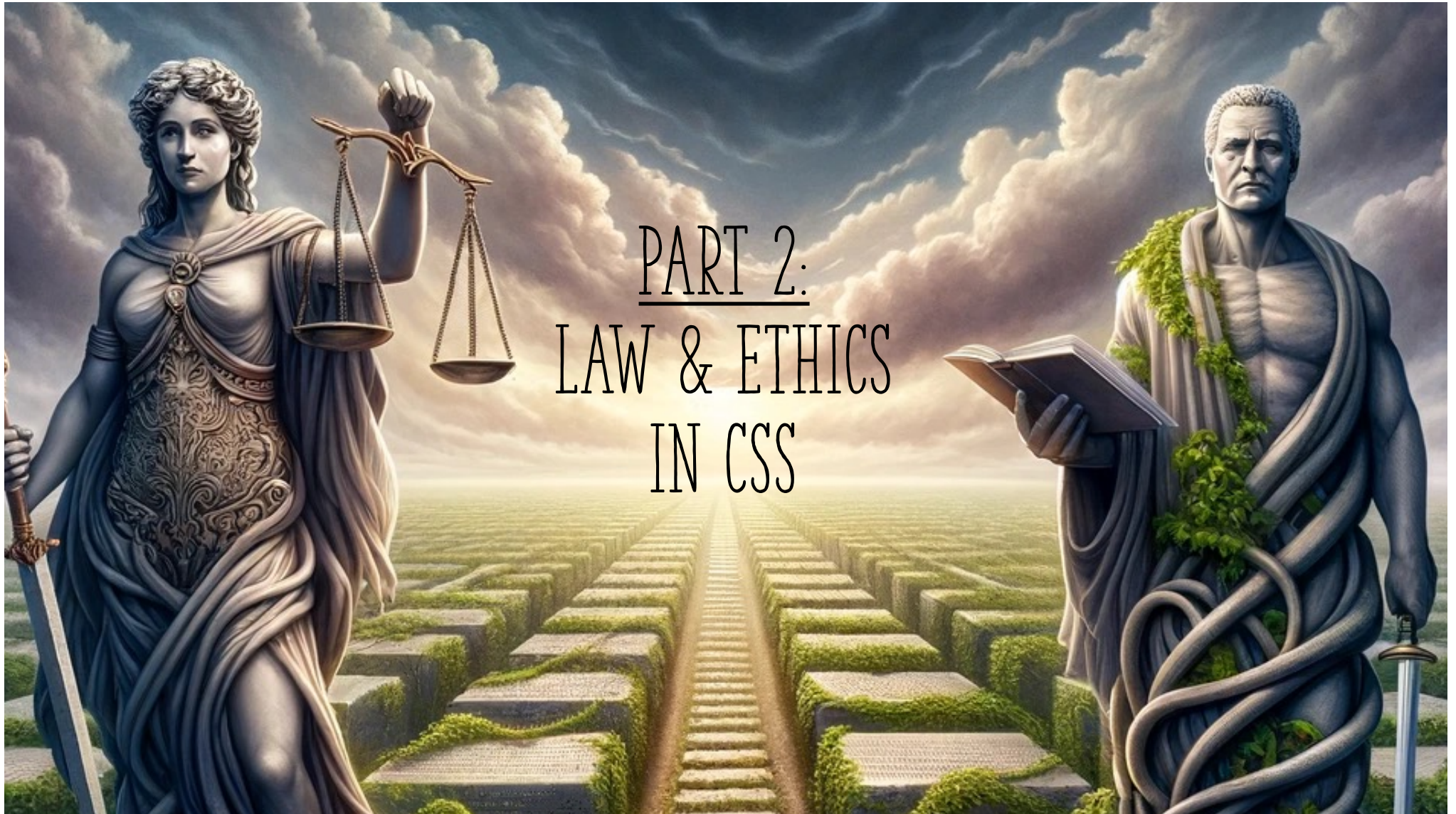
Is the data already available through an API or existing dataset?

Is the data source central to your contribution?

Do you have the time to do it well and legally/ethically?

Will data collection swallow the whole project?

Scraping is often very time-consuming, so best avoided for a 6-week project!



PART 2:
LAW & ETHICS
IN CSS

What's the relationship between Law and Ethics?



**FRIENDLY REMINDER:
LEGALITY ISN'T A GUIDE TO MORALITY**

ETHICS != LAW

- There are legal things that are not ethical.
- There are ethical things that are not legal.
- But we usually want to try to be *both* ethical and legal.



Concerns over data privacy and various scandals have meant more regulation

LAW

1. Terms of Service
2. Copyright Infringement
3. EU Database Directive in the EU
4. GDPR



TERMS OF SERVICE & SCRAPING

Can websites contractually limit scraping in their terms of use? *Yes, they can.*

But are those provisions enforceable? *Well, that depends. How was the contract created?*

- Browsewrap agreement: contracts that were concluded simply by visiting a website. Legal theory generally does not accept agreements of this type as valid.
- Clickwrap agreements: require an action by the user. "By pressing Accept, you agree to our Terms and Conditions". Clickwrap agreements are perfectly fine and fair, and the courts will readily enforce them.

Basically: If you need to log in, create a user, or click "I agree", you *have* signed a ToS and need to follow it.

If not, you have not entered a legal agreement, and you can *go wild!*

APIS AND TERMS OF SERVICE

- APIs often have ToS that you need to follow
- (In)famously, Twitter/X does not allow you to share any collected data
- Be mindful of the ToS when using APIs!

Developer terms

Developer Agreement and Policy

X Developer Agreement

Last Updated: November 14, 2023

By clicking "Accept & Subscribe", continuing to pay the recurring subscription fee for Paid Services, and by otherwise accessing or using any Licensed Material, you agree to the terms of our Developer Agreement. Subscriptions auto-renew until canceled, as described in the Terms. A verified phone number is required to subscribe. If you've subscribed on another platform, manage your subscription through that platform.

COPYRIGHT INFRINGEMENT

Scraping and reproducing content from a website could infringe on copyright laws.

In the EU, the scraping of copyrighted content is permitted (Article 3, 4 of Directive 2019/790 in the Digital Single Market DSM Directive).

- "any automated analytical technique aimed at analysing text and data in digital form in order to generate information which includes, but is not limited to, patterns, trends and correlations";

Scraping of copyrighted content is only permitted for the purposes of generating information.

For example, you can scrape a webpage to extract data to analyze using natural language analysis.

But you *cannot* scrape news articles and then republish them on your own website.

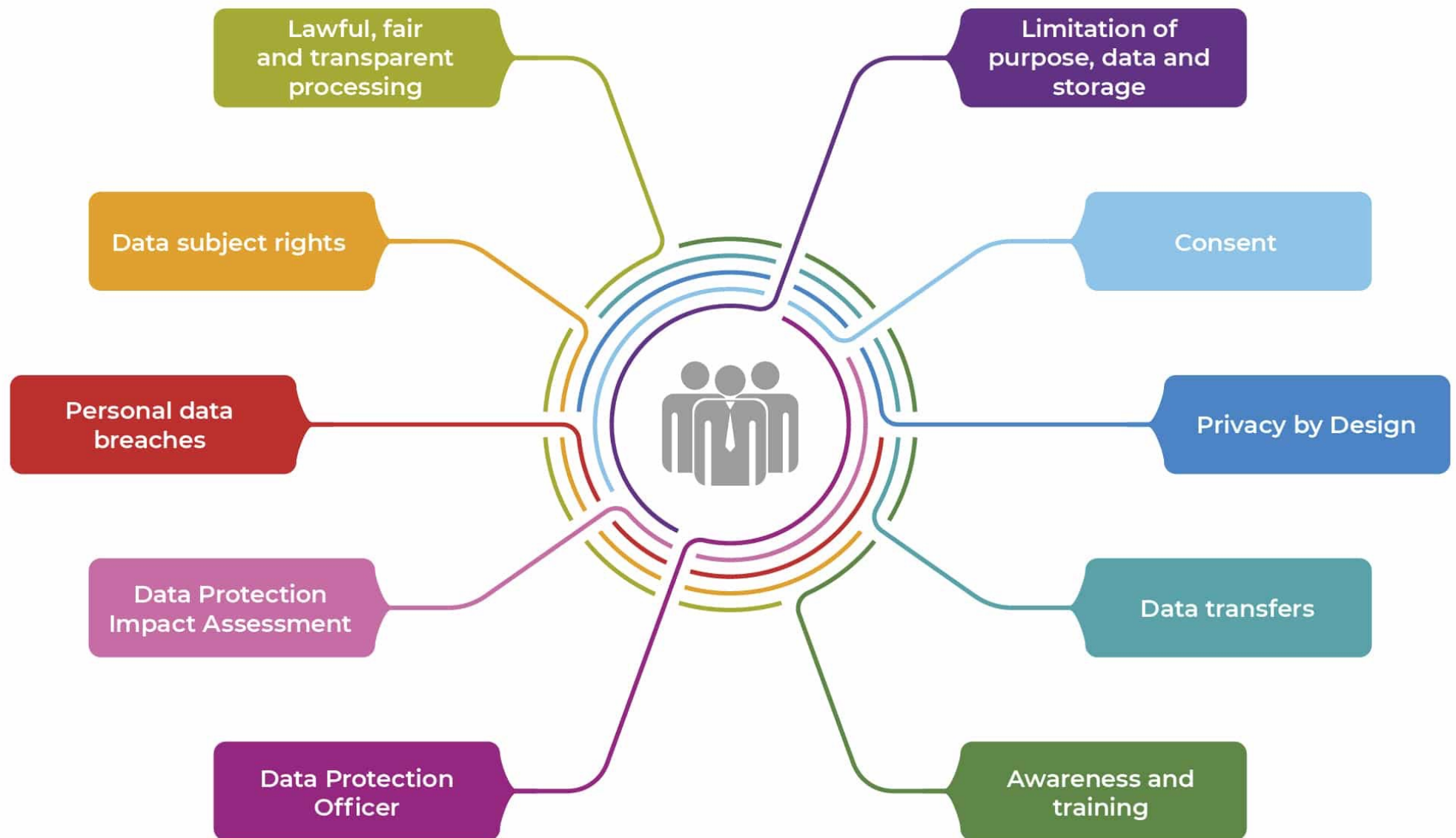
For scientific research, you can freely scrape almost anything! But do not republish original work, and be mindful of sharing scraped data publicly.



GDPR: GENERAL DATA PROTECTION REGULATION



- GDPR matters whenever you process personal data.
- The key questions are: What is the legal basis? What data do you really need? How will you inform people? How will you store and protect the data?
- Consent is one possible basis, but not the only one.





We value your privacy preferences

This website gathers information about how users interact with it, in order to continuously improve the service provided to you. Third party content may also be provided. Please confirm you're OK with this to continue.



Essential

Required for the website to function



Marketing

Measure the success of marketing campaigns



Analytics

Examine popular website areas, conversion rates, traffic sources, and browser details

ACCEPT ALL & CONTINUE

SAVE

You can change these settings at any time from our privacy policy page.

WHAT IS PERSONAL DATA ANYWAY?

GDPR only applies to personal data: "Personal data means any information relating to an identified or identifiable natural person"

If the data is not possible to link to a person, it does not fall under GDPR.

What does this actually mean? Debated.

E.g.. Name, surname, date of birth, address, social security number, passport number, national ID number, employment information, Contact details, phone number, email address, IP address, Facebook, Twitter, and other network handles, location either by address or GPS, shopping preferences, behavioral data, Video + audio recordings of people and biometric data. Special categories of personal data: sex, gender, and sexual orientation, racial or ethnic origin, religious beliefs, political opinions, medical records

DE-IDENTIFICATION BY ANONYMIZATION / PSEUDONYMIZATION

If research data are anonymised/pseudonymised, to what extent do they still fall under the GDPR?

- Anonymisation is the process of removing all the direct and indirect information that can link the data to an individual.

After anonymisation, nobody can link the data to an individual anymore, i.e. no key files, pseudonyms, etc. are used. When data is fully anonymised, the data isn't personal data anymore and it doesn't fall in the scope of the GDPR.

But re-identification is a constant risk!

PRINCIPLES OF GDPR

Data minimization: only data necessary should be processed, short storage period, limited accessibility

Privacy by design: any actions involving the processing of personal data is done with data protection and privacy in mind.

Privacy by default: all technical and organisational measures are taken to process data with the highest privacy protection.



GDPR PAPERWORK

1. Data management plan (DMP)
2. Privacy Protection Review (PPR)
3. Data Protection Impact Assessment (DPIA)
4. Processor agreement (PA)
5. Data exchange agreement (DEA)



ETHICS

1. Ethical frameworks
2. Ethical pluralism
3. AolR guidelines
4. ERB application
5. Ethics section in paper

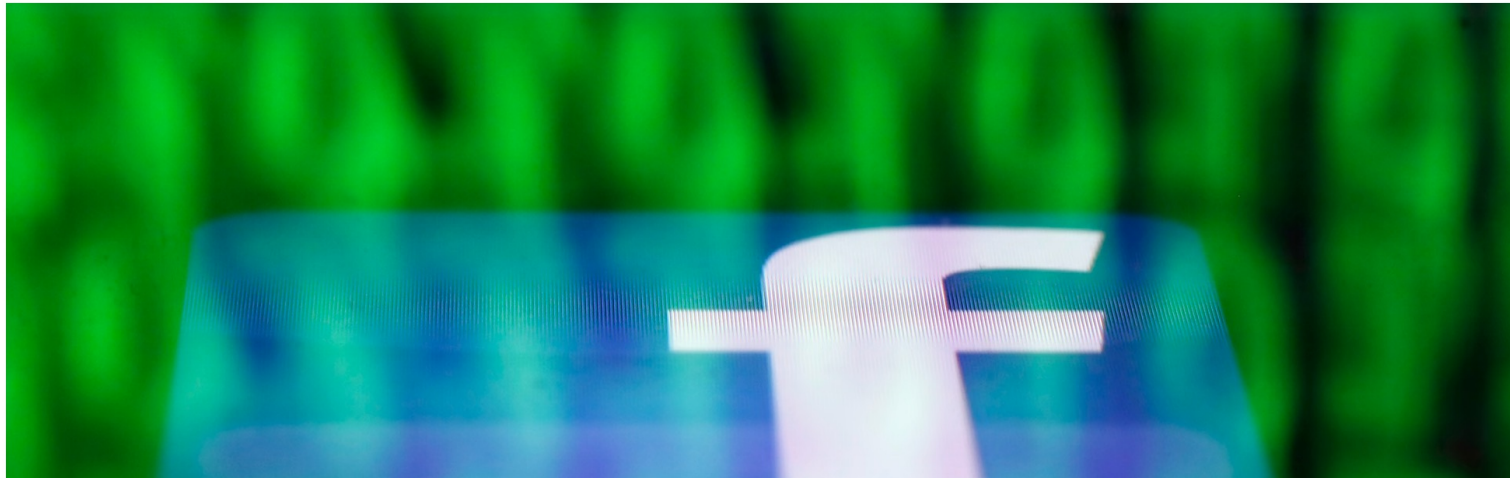
TECHNOLOGY

Everything We Know About Facebook's Secret Mood-Manipulation Experiment

It was probably legal. But was it ethical?

By Robinson Meyer

- 700,000 users were exposed either to negative posts or positive posts
 - Measured effects on their own posts
 - People got sadder
 - Published in PNAS
- Is this fine?



RESEARCH ETHICS IN CSS

- Social scientists have long had IRB: come from the tradition of experiments.
Stem from highly unethical experiments in the 1960s (e.g., Stanford Prison Experiment)
- Computer science has little tradition of minding ethics.

CSS is in the process of debating ethics. Still not well-established.

Varies depending on institution and region.

WHAT IS ETHICAL ANYWAY?

We can analyze our ethical dilemmas and challenges through a range of ethical frameworks:

Deontological: the rightness or wrongness of actions does not depend on their consequences but rather on whether they fulfill our duty or follow certain moral rules. Rule-based. Strong in Europe and Scandinavia.

Consequentialism: The outcomes and consequences of actions are what matter. (UK and US).

Utilitarianism: calculate the greater good for the collective and society in general.

Ethics of care: Prioritizes maintaining and fostering relationships, empathy, and caring for others, particularly those dependent on or vulnerable to one's actions.

So which one do we use?

AolR suggests ethical pluralism rather than a single master rule

ETHICAL PLURALISM AND CROSS-CULTURAL AWARENESS

Ethical pluralism: see the issue from multiple frameworks and discuss their competing views.

Accept ambiguity, uncertainty, and disagreement as inevitable.

Cross-cultural awareness: What are the expectations of the person being studied? What ethical framework do they follow?

If you're studying a distant culture, you must understand their views and expectations on ethics.

Research ethics is rarely black and white: there are competing interests that must be weighed

MAIN QUESTIONS

Who could be harmed, how, how likely is it, and what have you done to reduce that risk?

Are the benefits of the study large enough to warrant these potential harms?

DISCUSSION QUESTION: HARMING NAZIS



We generally want to avoid harming our research subjects.

But when studying Stormfront, the question was how fast to collect the data:

fast scraping means getting data in weeks instead of months, but may slow down their website.

But their website fuels neo-Nazi violence and has been linked to 100s of murders.

If we're harming self-proclaimed Nazis, does it make it more okay?

What would different ethical frameworks say?



ASSOCIATION
OF
INTERNET
RESEARCHERS

ETHICS GUIDELINES

Key Principles

- Respect for Persons: Emphasizes the importance of respecting the autonomy, privacy, and dignity of individuals and communities involved in or affected by the research.
- Beneficence: Encourages actions that contribute to the welfare of research participants, aiming to maximize benefits and minimize harm.
- Justice: Ensures fair treatment and equitable distribution of the benefits and burdens of research across different groups, avoiding exploitation.

WRITING AN ETHICS SECTION IN A PAPER

- Your paper should have an ethics section! Usually under "Methods & data"
- Describe your ethics process. Show that you've considered ethical implications.
- *Make an argument* for your work being ethical based on an existing framework or guidelines.
- *Very brief!*

We follow the ethical guidelines for Internet research provided by the Association of Internet Researchers ([Franzke et al., 2020](#)) and by the [British Sociological Association \(BSA, 2017\)](#). To ensure anonymity for the members of the community, usernames were deleted when collecting the corpus. Since the analytical focus is on broader discursive patterns, it is not possible to identify individual users from the results here presented. The project has been assessed and approved by the Regional Ethical Review Board.